PART-1 : PHYSICS

SECTION-I

1) A standing wave given by the equation $y = 2A \sin\left(\frac{\pi x}{L}\right) \sin \omega t$ is formed between the points x = –L and x = L. What is the minimum separation between two particles whose maximum velocity is half the maximum velocity of the antinodes?

(A) $\frac{L}{12}$

(B) $\frac{L}{6}$

(C) $\frac{L}{3}$

(D) $\frac{2L}{3}$

2) An object is placed at a distance of 20 cm from an equi-concave lens of focal length 30 cm whose one side is silvered. If a virtual image is formed at $\frac{20}{3}$ cm from the lens then what is the radius of curvature of the curved surface?

(A) 30 cm

(B) 40 cm

(C) 60 cm

(D) 15 cm

3) **Assertion :** Geostationary satellites may be setup in equatorial plane in orbits of any radius more than earth's radius.
Reason : Geostationary satellites have period of revolution of 24 hrs.

(A) Assertion is true, reason is true and reason is correct explanation for assertion.

(B) Assertion is true, reason is true and reason is NOT the correct explanation for assertion.

(C) Assertion is true, reason is false.

(D) Assertion is false, reason is true.

4) In a compound microscope, the focal length of objective lens is 1.2 cm and focal length of eye piece is 3.0 cm. When object is kept at 1.25 cm in front of objective, final image is formed at infinity. Magnifying power of the compound microscope should be (Least distance of distinct vision D = 25cm) :-

(A) 200

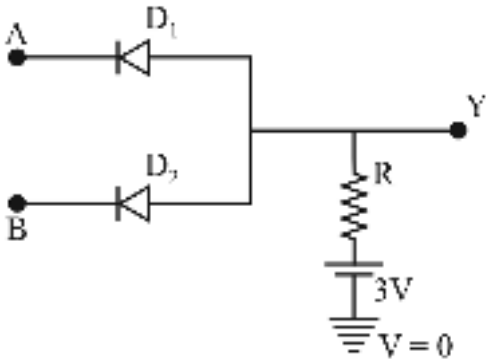
(B) 100

For More Material Join: @JEEAdvanced_2025

(C) 400

(D) 150

5) In the circuit, the logical value of A = 1 or B = 1 when potential at A or B is 3V and the logical



value of A = 0 or B = 0 when potential at A or B is 0 V.

(A)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

(B)

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

(C)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

(D)

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

6) Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 MeV. It splits in two fragments 'B' and 'C' of mass numbers 104 and 116. The binding energy of nucleons in 'B' and 'C' is 6.4 MeV per nucleon. The energy Q released per fission will be :-

For More Material Join: @JEEAdvanced_2025

(A) 0.8 MeV

(B) 275 MeV

(C) 220 MeV

(D) 176 MeV

7) A plane electromagnetic wave travels in a medium of relative permeability 1.6 and relative permittivity 6.4. If magnitude of magnetic intensity is 4.5×10^{-2} T at a point, what will be the approximate magnitude of electric field intensity at that point? (Give. permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}, speed of light in vacuum $c = 3 \times 10^8$ m/s)

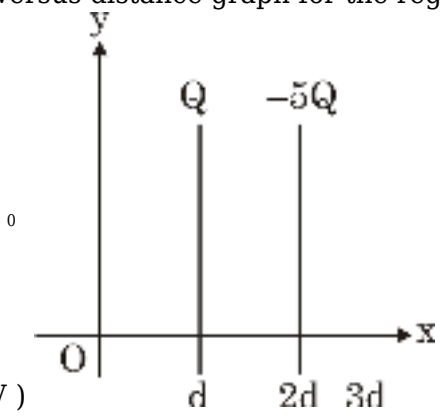
(A) 16.9 V/m

(B) 2.25×10^{-2} V/m

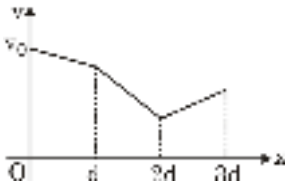
(C) 8.48 V/m

(D) 4.2×10^6 V/m

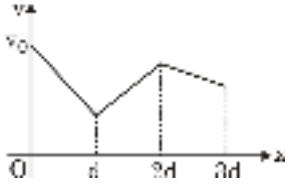
8) Two large identical plates are placed in front of each other at x = d and x = 2d as shown in the figure. If charges on plates are Q and –5Q, the potential versus distance graph for the region x = 0 and x = 3d is :- (d is very small and potential at x = 0 is V)



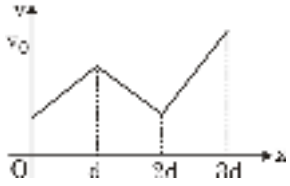
(A)



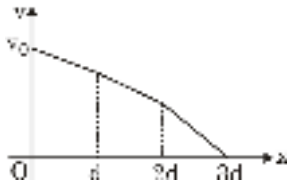
(B)



(C)

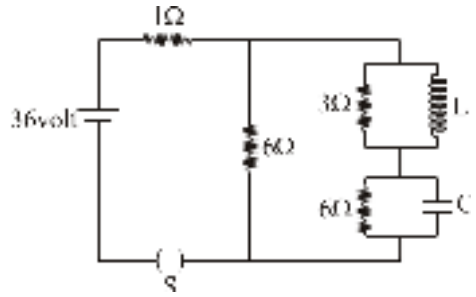


(D)



For More Material Join: @JEEAdvanced_2025

9)



List-I		List-II	
(P)	Current through battery just after closing of switch (in ampere)	(1)	12
(Q)	Current through capacitor just after closing of switch (in ampere)	(2)	8
(R)	Switch is closed and when steady state of current is achieved, the current through battery (in ampere)	(3)	9
(S)	After steady state switch is re-opened. Just after reopening of switch, the ratio of current through inductor and capacitor	(4)	1

(A) P - (1), Q - (2), R - (3), S - (4)

(B) P - (3), Q - (4), R - (2), S - (1)

(C) P - (4), Q - (3), R - (1), S - (2)

(D) P - (2), Q - (1), R - (4), S - (3)

10) A paramagnetic material has 10^{28} atom/m^3. Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4}. Its susceptibility at 300 K is :-

(A) 3.672×10^{-4}

(B) 3.726×10^{-4}

(C) 3.267×10^{-4}

(D) 2.672×10^{-4}

11) The value of γ is $\frac{4}{3}$ for an adiabatic process of an ideal gas for which internal energy U = k + nPV, where k is constant. The value of n is

(A) 2

(B) 3

(C) 4

(D) 5

12) A metallic wire of density d is lying horizontal on the surface of water (surface tension = T). The

For More Material Join: @JEEAdvanced_2025

maximum length of wire so that it may not sink will be :-

(A) $\sqrt{\frac{2Tg}{\pi d}}$

(B) $\sqrt{\frac{2\pi T}{dg}}$

(C) $\sqrt{\frac{2T}{\pi dg}}$

(D) any length

13) A black box is connected across an ac source of emf V = 100 sin(100 s^{-1}t) in volt. Phase of the current in the circuit is ahead of the phase of the source voltage by $\frac{\pi}{4}$. The peak current is observed to be $\sqrt{2}$A. The black box consists of :-

(A) Only a resistance of 50Ω.

(B) Only a capacitance of 0.2 mF.

(C) A series combination of a resistance of 50Ω and an inductance of 0.2 mH.

(D) A series combination of a resistance of 50Ω and a capacitance of 0.2 mF.

14) A and B are two bulbs with rated powers are 200V, 100W and 250V, 125W respectively. The bulbs are connected in parallel across the same dc source. The ratio of the power consumed by the bulb A to the bulb B is :-

(A) 4/5

(B) 5/4

(C) 2/5

(D) 5/2

15) **Statement-1 :** The electrostatic force between the plates of a charged isolated capacitor decreases when dielectric fills whole space between plates.
Statement-2 : The electric field between the plates of a charged isolated capacitor decreases when dielectric fills whole space between plates.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

16) In a young's double slit experiment light of wavelength 600 nm is used. P is the nearest point from the central maximum where intensity is 3/4th of the maximum intensity. What percentage of the maximum intensity is observed at point P if light is replaced by the wavelength 400 nm?

(A) 50%

(B) 75%

For More Material Join: @JEEAdvanced_2025

(C) 25%

(D) 100%

17) Two simple pendulum of length ℓ and 16 ℓ are released from the same phase together. They will be at the same phase after a minimum time :-

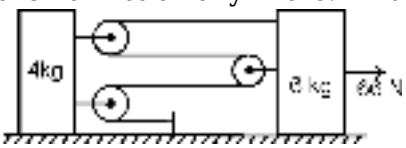
(A) $\frac{8\pi}{3}\sqrt{\frac{\ell}{g}}$

(B) $\frac{\pi}{3}\sqrt{\frac{\ell}{g}}$

(C) 2 s

(D) none of these

18) Consider the situation shown in the figure. A force F = 66 N is applied on 6 kg block. The pulleys and string are massless. There is no friction anywhere. Find the difference in acceleration of 6 kg block and 4 kg block ($a_6 - a_4$).



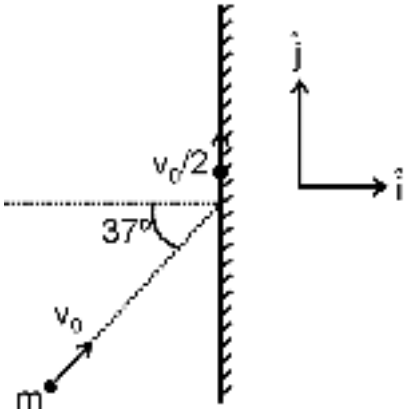
(A) 3 m/s^2

(B) – 3 m/s^2

(C) 2 m/s^2

(D) – 2 m/s^2

19) A ball collides with a wall and after collision moves parallel to the wall as shown in the figure.



The impulse acting on the ball during the collision is :-

(A) $mv_0\left(-\frac{4}{5}\hat{i}-\frac{1}{10}\hat{j}\right)$

(B) $mv_0\left(-\frac{4}{5}\hat{i}-\frac{1}{5}\hat{j}\right)$

(C) $mv_0\left(-\frac{3}{5}\hat{i}-\frac{1}{10}\hat{j}\right)$

(D) $mv_0\left(-\frac{3}{5}\hat{i}-\frac{1}{5}\hat{j}\right)$

For More Material Join: @JEEAdvanced_2025

20) A person is sitting on a cart which is accelerating with $10\sqrt{3}$ m/s^2. At a certain instant the person throws a ball vertically upwards with velocity $20\sqrt{3}$ m/s with respect to cart. Find the minimum radius of curvature in meter as observed by the person throwing the ball (take g = 10 m/s^2)

(A) 30 m

(B) $30\sqrt{2}$m

(C) 45 m

(D) $45\sqrt{2}$m

SECTION-II

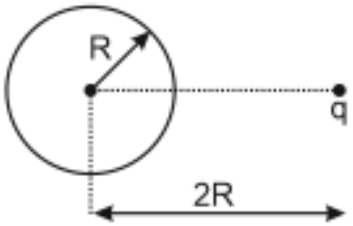
1) A hydrogen like atom of atomic number z is in excited state of quantum number 2n. It can emit a maximum energy photon of 204 eV. If makes a transition to the quantum state n, a photon of energy 40.8 eV is emitted. Calculate the atomic number z. Ground state energy of hydrogen atom is –13.6 eV.

2) A solenoid of length of 0.4 m and having 500 turns of wire carries a current of 3 ampere. A thin circular coil having 10 turns of wire and of radius 0.01 m carries a current of 4 ampere. Calculate the torque (in μNm) required to hold the coil in the middle of the solenoid with its axis perpendicular to the axis of the solenoid. (Take $\pi^2 = 10$)

3) A charged particle having specific charge ratio α CKg^{-1} leaves the origin with a velocity $v_0\hat{i}$. $\vec{E}$ and $\vec{B}$ uniform electric field and magnetic field are directed along y-axis. The value of y-coordinate of the particle when it crosses the y-axis third time is $\frac{x\pi^2 E}{\alpha B^2}$. The value of x is :-

4) The power radiated by a black body is P, and it radiates maximum energy at the wavelength λ_0. If the temperature of black body is now changed so that it radiates maximum energy at wavelength $\frac{3\lambda_0}{4}$, the power radiated by it is $\frac{nP}{81}$. Then find the value of 'n'.

5) A point charge is placed outside a uncharged conducting spherical shell as shown in the figure. Find the electric potential (in kilo volt) on the surface of shell :- (Take : q = 1 μc and R = 50 cm)

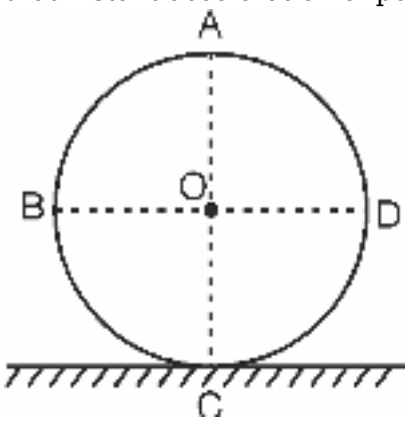


6) A block of mass 1 kg is released from rest on an inclined plane of inclination 30º. The block acquires the kinetic energy of 5J when slides down the plane by 2 m. How much force (in newton) is needed to be applied on the block to move up the plane so that it will gain the same kinetic energy over same path length ? Take g = 10 m/s^2.

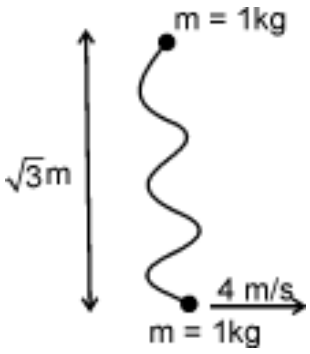
For More Material Join: @JEEAdvanced_2025

7) A sonometer wire resonates with frequency f_0 in its fundamental mode. If length of the wire is decreased to the value 0.95 m, the resonating frequency increases by 5 Hz and if length is increased to the value 1.0 m, the resonating frequency decreases by 5 Hz. Find the length (in cm) of the wire for the resonating frequency f_0. The speed of the wave in the wire is 382.2 m/s.

8) A wheel rolls on a plane surface without slipping. At a certain instant velocity and acceleration of centre 'O' are 1m/s and 3m/s^2. The radius of wheel is 0.5 m. BD is horizontal and AC is vertical. If at that instant acceleration of point A is $\sqrt{a}$ m/s^2, then find the value of 'a' :-



9) A physical quantity p is given as $p = \frac{a^2b}{(4-c)^3}$, where a = (2 ± 0.01) in SI units, b = (4 ± 0.02) in SI units and c = (2 ± 0.01) in SI units. Find percentage error in calculation of p.



10) Two particles each of mass 1 kg are connected by a string of length 2 m and placed on a smooth horizontal plane. Initially they are separated by $\sqrt{3}$ m. If one of the particle is given velocity 4 m/s as shown in the figure, find the tension in string in newton just after it becomes taut.

PART-2 : CHEMISTRY

SECTION-I

1) Consider the following reactions :
$IO_3^- + 6H^+ + 6e^- \longrightarrow I^- + 3H_2O$; $E° = +0.907$ V
$IO_3^- + 3H_2O + 6e^- \longrightarrow I^- + 6OH^-$; $E° = +0.260$ V
Predict where does IO_3^- acts as a better oxidant ?

For More Material Join: @JEEAdvanced_2025

(A) In acidic medium

(B) In basic medium

(C) In pure water

(D) Equally effective in both acidic and basic mediums

2) One mole of helium (He) gas is added to the equilibrium $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ set up in cylinder fitted with piston in such a way that the piston is moved outwards to keep the total equilibrium pressure constant :

(A) The equilibrium will remains unchanged

(B) The equilibrium will shift in the forward direction

(C) The equilibrium will shift in the backward direction

(D) None of these

3) $S(s) + \frac{3}{2}O_2(g) \longrightarrow SO_3(g) + 2x$ kcal

$SO_2(g) + \frac{1}{2}O_2(g) \longrightarrow SO_3(g) + y$ kcal

Find out the heat of formation of $SO_2(g)$:

(A) (y – 2x)

(B) (2x + y)

(C) (x + y)

(D) 2x/y

4) The number of moles of NaOH required to completely react with 20 ml of 0.1 M NaH_2PO_3 solution and 40 ml of 0.1 M $NaHCO_3$:

(A) 8×10^{-3}

(B) 2×10^{-3}

(C) 4×10^{-3}

(D) 6×10^{-3}

5) **Statement-I :** A reaction which is spontaneous and accompained by decrease of randomness must be exothermic.
Statement-II : All exothermic reactions are accompained by decrease of randomness.

(A) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.

(B) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.

(C) Statement-I is true, statement-II is false.

(D) Statement-I is false, statement-II is true.

6) Match the column for the reaction of type $A(g) \longrightarrow 2B(g)$

Column-I	Column-II

For More Material Join: @JEEAdvanced_2025

(a)	[A] vs t for zero order	(p)	
(b)	$\frac{-d[A]}{dt}$ vs t for first order	(q)	
(c)	[B] vs t for first order	(r)	
(d)	$\frac{d[B]}{dt}$ vs $\frac{-d[A]}{dt}$ for first order	(s)	

(A) a-r, b-s, c-q, d-p

(B) a-q, b-r, c-p, d-s

(C) a-s, b-r, c-q, d-p

(D) a-q, b-s, c-r, d-p

7) Calculate the pH of a solution containing 0.2 M HCO_3^- and 0.1 M CO_3^{2-} :
[$K_1(H_2CO_3) = 4 \times 10^{-7}$; $K_2(HCO_3^-) = 4 \times 10^{-11}$]

(A) 10.4

(B) 10.1

(C) 6.1

(D) 10.7

8) In which reaction reddish brown vapour is evolved ?

(A) $CrO_4^{2-} + H_2SO_{4(conc.)} + H_2O_2 \longrightarrow$

(B) $Mn^{2+} + (NH_4)_2S_2O_8 + H^+ \longrightarrow$

(C) $K_2Cr_2O_{7(s)} + \text{conc. } H_2SO_4 + Cl^- \xrightarrow{\Delta}$

(D) $MnO_2 + HCl \longrightarrow$

9) Select the incorrect statement with respect to oxides and oxoanions of transition metals.

(A) Among oxides of chromium, CrO is basic, Cr_2O_3 is amphoteric and CrO_3 is acidic.

(B) No higher oxides of iron above Fe_2O_3 are known.

For More Material Join: @JEEAdvanced_2025

(C) Ti,V, Cr and Mn form oxides MO and their correct increasing order of acidic character is MnO < CrO < VO < TiO.

(D) Vanadium(V) oxide react with acids as well as alkalies.

10)

Which of the following statement is **INCORRECT** ?

(A) Np is one of the trans-uranium element

(B) Pm is the only synthetic radioactive lanthanide

(C) LnC_2 on hydrolysis gives C_2H_2

(D) Actinide contraction is result of poor shielding of 4f-electrons.

11) $2NaX(aq) + \text{conc. } H_2SO_4 \rightarrow 2HX + Na_2SO_4$
$HX + \text{Carbon} \rightarrow$ Three oxides are formed.
X is an unknown anion
The correct statement is :-

(A) X can be F, Cl , Br or I

(B) All the oxides formed are acidic in nature

(C) Two of the oxides formed are gases

(D) HX is a good reducing agent

12) Which is the correct order of bond angle ?

(A) $NH_3 < NF_3 < OCl_2$

(B) $PH_3 < SbH_3 < NH_3$

(C) $H_2O > Cl_2O > OF_2$

(D) $H_2Te < H_2Se < H_2S$

13) A cobalt amine has the formula $Co(H_2O)_6Cl_3$, with $AgNO_3$ solution, one-third of the chloride is precipitated. The complex shows cis-trans isomerism. It can have the structure :

(A) $[Co(H_2O)_6]Cl_3$

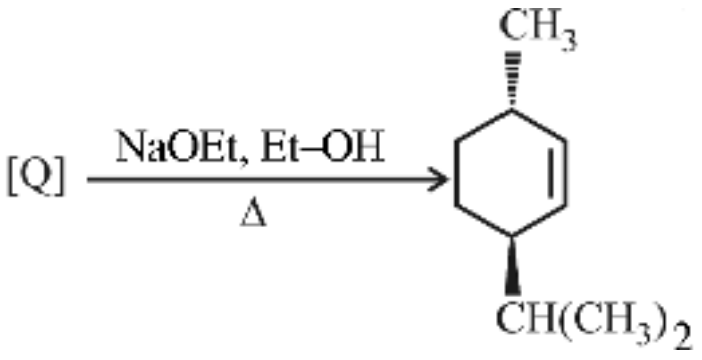
(B) $[Co(H_2O)_5Cl]Cl_2.H_2O$

(C) $[Co(H_2O)_4Cl_2]Cl.2H_2O$

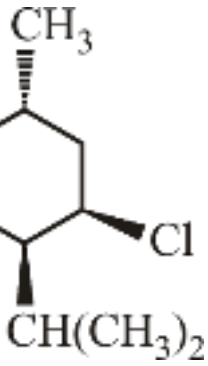
(D) $[Co(NH_3)_2 \cdot (H_2O)_2Cl_2]Cl.4H_2O$

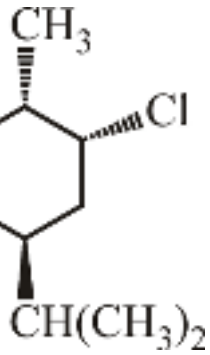
14) In the following reaction reactant(s) [Q] is

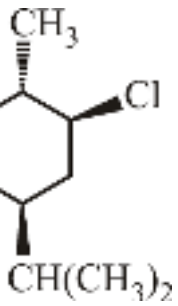
For More Material Join: @JEEAdvanced_2025

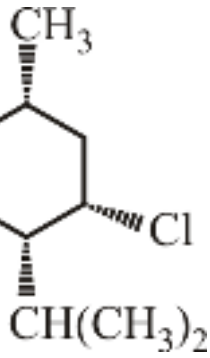


(Only product)

(A)


(B)


(C)


(D)


15) Correct statement among the following is-

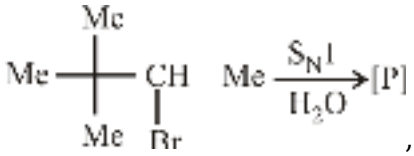
(A) Proline is an example of aromatic heterocyclic ring containing amino acid.

(B) α-D-⊕-glucopyranose and β-D-⊕-glucopyranose differ in their configuration at C_2- position.

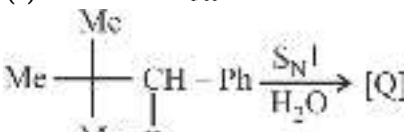
For More Material Join: @JEEAdvanced_2025

(C) The pKa- value of aspartic acid is higher than glutamic-acid.

(D) The pentaacetate of glucose does not react with hydroxylamine.

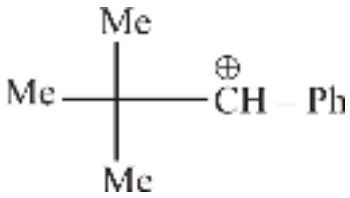
16) (I) 

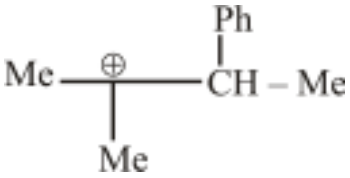
, Major

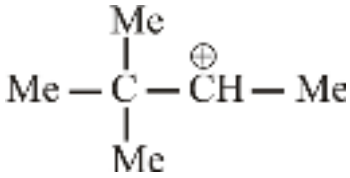
(II) 

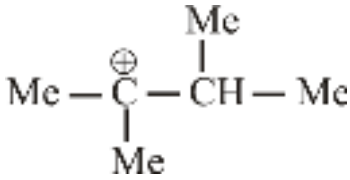
, Major

Identify the incorrect intermediate carbocation which is not formed during the reaction (I) and (II)?

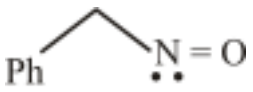
(A) 

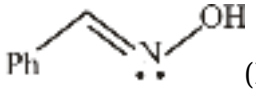
(B) 

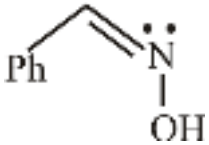
(C) 

(D) 

17) The incorrect statements about the following species is ?

(I) 

(II) 

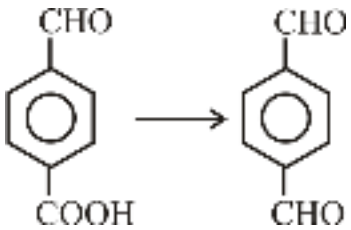
(III) 

(A) (I) is more stable than (II) and (III)

(B) (II) and (III) are more stable than (I)

(C) (II) and (III) are diastereomers

(D) (III) is a toutomer of (I)

18) 

For More Material Join: @JEEAdvanced_2025

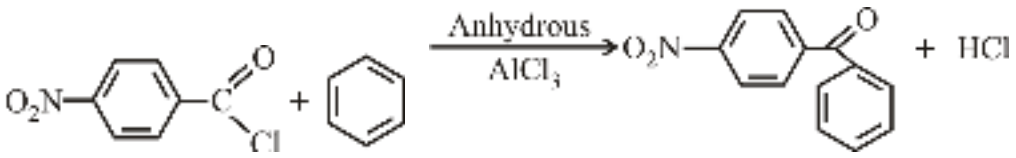
(A) conc. HI/Red-P

(B) (i) PCl_5 (ii) H_2 , Ni, Δ

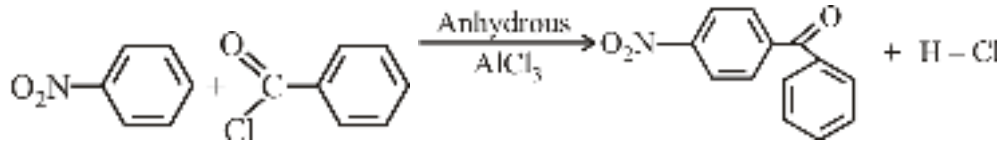
(C) (i) ethylene glycol/ H^+ (ii) $SOCl_2$
(iii) $H_2/Pd.BaSO_4$ (iv) H_3O^+

(D) (i) ethylene glycol, N_2H_4 , OH^- (ii) B_2H_6

19) Path-I



Path-II



The correct statement among the following about the above two paths of synthesizing the organic product P is :

(A) Path I is better than Path II

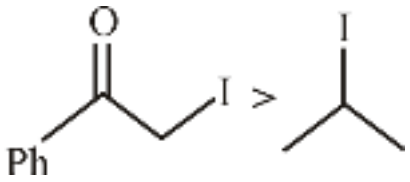
(B) Path II is better than Path I

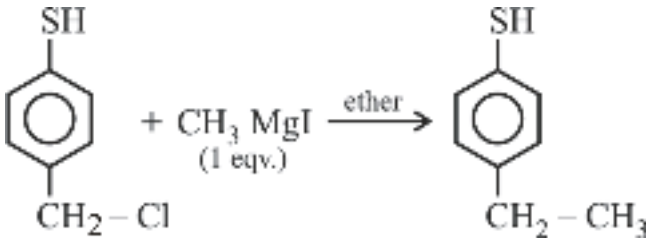
(C) Both the paths are equally good

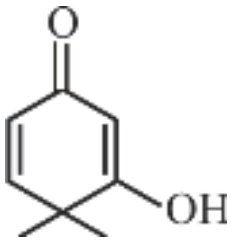
(D) Path I will give less yield of the product as compared to that obtained by path II

20) Choose the incorrect option :

(A) Picric acid and benzyl alcohol can be distinguish by $NaHCO_3$

(B)  rate of S_N2 reaction

(C) 

(D)  show tautomerism.

SECTION-II

1) Calcium oxide (CaO) prepared by reaction
$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
Calculate free energy change for the reaction ($\Delta_rG°$) in kJ/mole at 300 K

For More Material Join: @JEEAdvanced_2025

Given :
$\Delta_r H°(CaO, s) = -635$ kJ/mole ; $\Delta_f H°(CO_2, g) = -394$ kJ/mole
$\Delta_f H°(CaCO_3, s) = -1207$ kJ/mole
$S°(CO_2, g) = 214$ J/K-mole ; $S°(CaO, s) = 40$ J/K-mole
$S°(CaCO_3, s) = 94$ J/K-mole; all given data at 300 K.

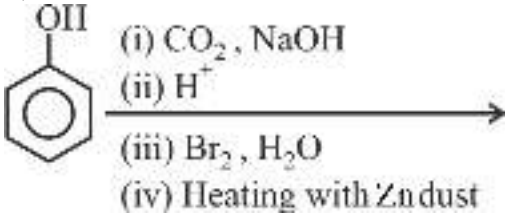
2) Reaction : $CH_3CHO + 2H^+ + 2e^- \rightarrow C_2H_5OH$ $E° = 1.76$ V
Calculate E at pH = 1 and 298 K
Given : $[C_2H_5OH] = 5 \times 10^{-10}$ M and $[CH_3CHO] = 5$M
$\left[\text{Take : } \frac{2.303\,RT}{F} = 0.06\right]$

3) Find the amount (in gm) of sample containing 80% NaOH, required to prepare 60 litre of 0.5 M solution.

4) How many of the following can show colour due to charge transfer spectra?
$KMnO_4$, K_2CrO_4, $K_2Cr_2O_7$, $KAlO_2$, K_2CO_3, KCl, K_2SO_4 , Cl_2 , $FeSO_4.7H_2O$

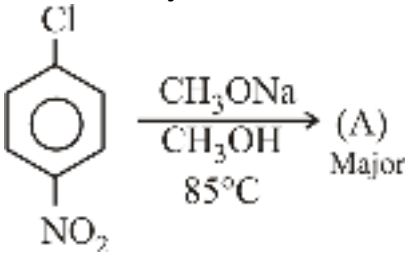
5) Consider the following phosphorus based oxoacids
$H_4P_2O_6$, $H_3P_3O_9$, $H_4P_2O_8$, $H_4P_2O_5$ and $H_5P_3O_{10}$
Amongst these oxoacids, the number of oxoacids with P–O–P bond is _____ :

6) Find the sum of locant of Br in final product as per the IUPAC nomenclature.



7) 2.12 gram of benzaldehyde was reacted with acetone in presence of C_2H_2ONa to give 1.46 gram of the enone product. Find percentage yield of the reaction is :

8) How many SP^2 carbon atom(s) present in major product (A) in given reaction?



9) When XO_2 is fused with an alkali metal hydroxide in presence of an oxidizing agent such as KNO_3 ; a dark green product is formed which disproportionates in acidic solution to produce a dark purple solution, the number of d-orbitals used in hybridisation central atom of dark green product formed.

For More Material Join: @JEEAdvanced_2025

10) Find the sum of lone pairs on central atom of following species.
SiF_4 , SF_4 , SF_2 , SF_6 , H_2S , SO_2 , SO_3

PART-3 : MATHEMATICS

SECTION-I

1) Let $\alpha_1, \alpha_2, ..., \alpha_8$ be roots of the equation $1 + z + z^2 + ... + z^8 = 0$. On throwing three dice simultaneously, sum of the numbers appearing on them is n. The probability such that $\sum_{i=1}^{8} (\alpha_i)^n = 8$ is $\frac{a}{b}$ (where a and b are co-prime) then the value $|7a - b|$ is :

(A) 22
(B) 16
(C) 26
(D) 17

2) Area bounded by $-8z^2 - 8\bar{z}^2 + 20z\bar{z} = 144$ is (where z is a complex number) :

(A) π
(B) 6π
(C) 12π
(D) 4π

3) **Assertion A :** The number of ways in which 3 married couples with their 4 children can sit in a row such that no husband and wife are together, are 10! – 3. 9! + 3 . 8! – 7!.
Reason R : Number of ways of occurrence of at least one event out of three events A, B & C = n(A ∪ B ∪ C) = n(A) + n(B) + n(C) – n(A ∩ B) – n(B ∩ C) – n(C ∩ A) +n(A ∩ B ∩ C).

(A) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**
(B) Both **A** and **R** are correct and **R** is the correct explanation of **A**
(C) **A** is correct but **R** is not correct
(D) **A** is not correct but **R** is correct

4) Let $\vec{a}.\vec{b} = 7$, $\vec{b}.\vec{c} = 21$ and $\vec{c}.\vec{a} = 9$, then volume of parallelopiped determined by the vectors $\vec{a} + (\vec{a} \times \vec{b}) \times \vec{c}$, $(\vec{b} \times \vec{c}) \times \vec{a}$ and $(\vec{c} \times \vec{a}) \times \vec{b}$ is (where $a^2 = 9$, $b^2 = 30$, $c^2 = 18$)

(A) $189\sqrt{218}$
(B) 945
(C) $63\sqrt{218}$
(D) None of these

5) Normal at P, Q, R drawn to $y^2 = 4x$ which intersect at (3, 0), then ΔPQR is :

For More Material Join: @JEEAdvanced_2025

(A) acute angled but not equilateral

(B) obtuse angled triangle

(C) equilateral triangle

(D) scalene triangle

6) If $f(x) = \left(x^3 - x^2 - 16x + 16\right)\left|\left(x^2 + 2x - 3\right)\ell nx\right|$, then number of points where $f(x)$ is non-derivable is :

(A) 0

(B) 1

(C) 2

(D) 3

7) If function $f(x) = (x^2 - 3x + m)\text{sgn}(x^2 - 2x + 2m)$ is discontinuous at exactly one point then number of possible values of m is :-
(where sgn( ) denotes signum function)

(A) 0

(B) 1

(C) 2

(D) 3

8) If eccentricity of conjugate hyperbola of given hyperbola $\left|\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2}\right| = 3$ is e', then the value of 8e' is :

(A) 5

(B) 8

(C) 10

(D) 12

9) Match List-I with List-II and select the correct answer using the code given below the list.

List-I		**List-II**	
(A)	Let $\left(\vec{a} \times \vec{b}\right) \times \vec{c} = -5\vec{a} + 4\vec{b}$ and $\vec{a}.\vec{b} = 3$ and $\vec{a} \times \left(\vec{b} \times \vec{c}\right) = \lambda\vec{b} + \mu\vec{c}$, then $\lvert\lambda\rvert + \lvert\mu\rvert$ is (Where $\vec{a}$, $\vec{b}$, $\vec{c}$ are pairwise non-collinear)	(I)	2

For More Material Join: @JEEAdvanced_2025

(B)	If the line $y-\sqrt{3}x+3=0$ cuts the parabola $y^2 = x + 2$ at A and B, given that $\lvert(PA)\,(PB)\rvert = \frac{\lambda_1\left(2+\sqrt{3}\right)}{\lambda_2}$, then $\frac{\lambda_1^2+\lambda_2^2}{5}$ is (where P is $\left(\sqrt{3},0\right)$ and λ_1, λ_2 are relatively prime)	(II)	7
(C)	The sum (in radians) of all values of x with $0 \le x \le 2\pi$ which satisfy $\sqrt{2}(\cos 2x - \sin x - 1) = 1 + 2 \sin x$, is $k\pi$, then k is	(III)	6
(D)	Value of $\int_{-\pi/4}^{\pi/4} \frac{x^9-3x^7+x^5+1}{\cos^2 x}dx$ is equal to	(IV)	5

(A) (A) – I, (B) – II, (C) – III, (D) – IV

(B) (A) – II, (B) – IV, (C) – III, (D) – I

(C) (A) – IV, (B) – I, (C) – III, (D) – II

(D) (A) – III, (B) – IV, (C) – II, (D) – I

10) If m is the slope of a line which is tangent to $y^3 = x^4$ & a normal to $x^2 - 2x + y^2 = 0$, then $\left(\frac{3m}{4}\right)^3$ is equal to ($m \neq 0$) :

(A) 3

(B) $\frac{4}{3}$

(C) 4

(D) $\frac{3}{4}$

11) If $\int \frac{(\sqrt{x})^5}{(\sqrt{x})^7+x^6}dx = \frac{1}{k}\ln\left(\frac{x^k}{1+x^k}\right)+c$ then k is equal to :

(A) $\frac{1}{5}$

(B) $\frac{2}{5}$

(C) 5

(D) $\frac{5}{2}$

12) The area of the region which consists of all the points satisfying the conditions $|x - y|+|x + y| \le 8$, $xy \ge 2$, $x \ge 0$ and $y \ge 0$ is equal to -

(A) 14 – □n64

(B) 28 – □n64

For More Material Join: @JEEAdvanced_2025

(C) 14 + □n4

(D) 18 – □n4

13) Arithemetic Mean of observations $({}^nC_0)^2, ({}^nC_1)^2, ({}^nC_2)^2 \ldots ({}^nC_n)^2$ will be :

(A) $\frac{\lfloor 2n}{\lfloor n \ \lfloor n}$

(B) $\frac{\lfloor 2n}{(\lfloor n)\ 3}$

(C) $\frac{\lfloor 2n}{\lfloor n \ \lfloor n+1}$

(D) $\frac{\lfloor 2n}{\lfloor n \ (\lfloor n+1)^2}$

14) If $\sum_{i=0}^{2n} \lambda_i (a-2)^i = \sum_{i=0}^{2n} \mu_i (a-3)^i$ and $\lambda_i = 1 \ \forall \ i \geq n$ then :-

(A) $\mu_n = {}^{2n+1}C_n$

(B) $\mu_n = {}^{2n+1}C_3$

(C) $\mu_n = {}^{2n+1}C_n + {}^{2n+1}C_{2n}$

(D) $\mu_n = \sum_{i=n+1}^{2n} \lambda_i$

15) **Statement-1 :** If a + b + c + d = 9 and $a^2 + b^2 + c^2 + d^2 = 27$, where a, b, c, d are non-negative real numbers, then $d \in \left[0, \frac{9}{2}\right]$.

Statement-2 : $\left(\frac{a+b+c}{3}\right)^2 \leqslant \frac{a^2+b^2+c^2}{3}$.

(A) Statement-1 is True, Statement-2 is True

(B) Statement-1 is False, Statement-2 is False

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

16) f (x) is a diffrentiable function satisfy the relationship $f^2(x) + f^2(y) + 2(xy - 1) = f^2(x + y) \ \forall \ x, y \in$ R. Also f (x) > 0 $\forall$ x ∈ R , and $f\left(\sqrt{2}\right) = 2$. then $f\left(\sqrt{7}\right) =$

(A) 3

(B) 4

(C) 5

(D) 7

For More Material Join: @JEEAdvanced_2025

17) Number of roots of the function f (x) = $\frac{1}{(x+1)^3}$ – 3x + sin x is :

(A) 0

(B) 1

(C) 2

(D) More than 2

18) Let f (x) = $\int_3^x \frac{dt}{\sqrt{t^4+3t^2+13}}$. If g (x) is the inverse of f (x) then g'(0) has the value equal to :

(A) $\frac{1}{11}$

(B) 11

(C) $\sqrt{13}$

(D) $\frac{1}{\sqrt{13}}$

19) The value of the expression, $14\tan\left(\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{13}+\tan^{-1}\frac{1}{21}+\tan^{-1}\frac{1}{31}\right)$ is an integer which is equal to :

(A) 2

(B) 5

(C) 7

(D) 10

20) $\lim_{x\to 0} \frac{(e^{2x^2}-1-2x^2)(\cos x-1)}{(\sin 3x-\ln(1+3x))\ x^4}$:

(A) $\frac{2}{9}$

(B) $\frac{1}{9}$

(C) $-\frac{1}{9}$

(D) $-\frac{2}{9}$

SECTION-II

1) From any point on line (t + 2)(x + y) = 1, t ≠ –2 tangents are drawn to ellipse $4x^2 + 16y^2 = 1$. If chord of contact passes through a fixed point, then number of integral values of t for which fixed point lies inside ellipse is :

For More Material Join: @JEEAdvanced_2025

2) Let $A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 1 \end{bmatrix}$ then absolute value of $\frac{|(A-I)^3 - 5A|}{8}$ is :

3) Let $f(x) = x(x^2 - 1)(x^2 - 4)(x^2 - 9)$. Let α_i, i = 1,2,3....6 are the root of $f'(x) = 0$. Then the value of $\sum_{i=1}^{6} (-1)^i [\alpha_i]$, is (where [.] denotes greatest integer function and $\alpha_1 < \alpha_2 < \alpha_3 < \alpha_4 < \alpha_5 < \alpha_6$) :

4) A straight line drawn through the point P(–1, 2) meets the hyperbola $xy = c^2$ at the points A & B (Point A & B lie on same side of P). Let Point Q is collinear with A & B such that PA, PQ & PB are in A.P. If locus of Q is $y = \frac{\lambda_1 x}{(1 + \lambda_2 x)}$, then $\lambda_1 + \lambda_2$ is equal to :

5) Consider the quadratic polynomial $f(x) = x^2 - px + q$ where $f(x) = 0$ has prime roots. If $p + q = 11$ and $a = p^2 + q^2$, then find the value of $f(a)$ (where a is an odd positive integer).

6) If the equation of the circle whose radius is 3 and which touches the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ internally at the point (–1, – 1) is $ax^2 + by^2 + 2gx + 2fy - 32 = 0$, then value of $a + b - g - f = ?$

7) Suppose $f(x) = x^3 + ax^2 + bx + c$ satisfies $f(-2) = -10$ and takes the extreme value $\frac{50}{27}$ where $x = \frac{2}{3}$, then the value of $a + b + c$ is :

8) Let $I = \int_0^{\pi/4} (\pi x - 4x^2) \ln(1 + \tan x) dx$. If the value of $I = \frac{\pi^3 \ln 2}{k}$ where $k \in N$, find k.

9) Suppose f and g are two functions such that f, g : R → R, $f(x) = \ln\left(1 + \sqrt{1 + x^2}\right)$ and $g(x) = \ln\left(x + \sqrt{1 + x^2}\right)$ then value of $xe^{g(x)}\left(f\left(\frac{1}{x}\right)\right)' + g'(x)$ at x = 1 is :

10) Let $\tan\alpha \cdot \tan\beta = \frac{1}{\sqrt{2005}}$. Find the value of $(1003 - 1002\cos 2\alpha)(1003 - 1002\cos 2\beta)$:

For More Material Join: @JEEAdvanced_2025

ANSWER KEYS

PART-1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	C	C	D	A	A	D	D	A	A	C	B	D	D	B	D	A	A	C	A	C

SECTION-II

Q.	21	22	23	24	25	26	27	28	29	30
A.	4	60	18	256	9	10	98	40	3	3

PART-2 : CHEMISTRY

SECTION-I

Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	A	B	A	D	C	B	B	C	C	D	C	D	C	C	D	B	A	C	A	C

SECTION-II

Q.	51	52	53	54	55	56	57	58	59	60
A.	130	2	1500	3	3	9	50	6	3	6

PART-3 : MATHEMATICS

SECTION-I

Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	D	C	D	B	B	A	D	C	B	C	D	A	C	A	A	A	C	B	D	D

SECTION-II

Q.	81	82	83	84	85	86	87	88	89	90
A.	2	8	3	4	3422	21	1	192	0	2005

For More Material Join: @JEEAdvanced_2025

SOLUTIONS

PART-1 : PHYSICS

1) $\frac{\partial y}{\partial t} = 2A\omega \sin\left(\frac{\pi x}{L}\right)\cos\omega t$

$\frac{\partial y}{\partial t} = 2A\omega \sin\left(\frac{\pi x}{L}\right) = A\omega$

$\sin\left(\frac{\pi x}{L}\right) = \frac{1}{2}$

Separation, $\Delta x = L/3$

2) Now, $\frac{1}{f_{eff}} = \frac{1}{f_m} - \frac{2}{f_0}$

$\frac{1}{10} = \frac{1}{f_m} + \frac{2}{30}$

$\frac{1}{f_m} = \frac{1}{10} - \frac{2}{30} = \frac{1}{30}$

$f_m = +30$ cm

R = +60 cm

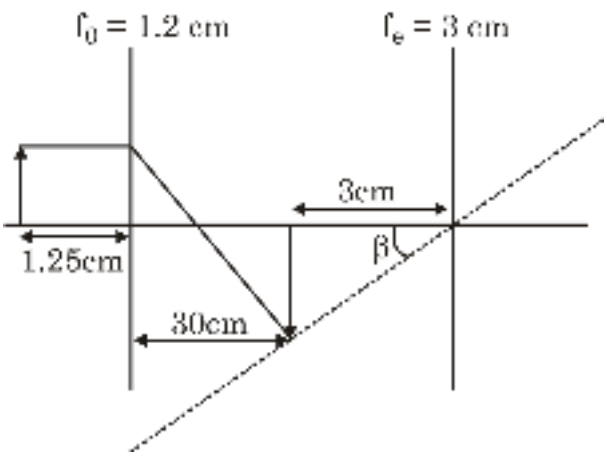
4) $f_0 = 1.2$ cm, $f_e = 3$ cm

$\frac{1}{f_0} = \frac{1}{v_0} + \frac{1}{u_0}$

$\frac{1}{1.2} = \frac{1}{v_0} - \frac{1}{(-1.25)}$

$\Rightarrow \frac{1}{v_0} = \frac{5}{6} - \frac{4}{5} = \frac{1}{30}$

M.P.$= \left(\frac{v_0}{u_0}\right)\left(\frac{D}{f_e}\right) = \left(\frac{30}{1.25}\right)\left(\frac{25}{3}\right) = 200$



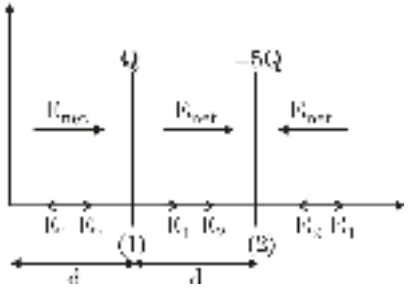
5) When any input is '0' (zero) then current flows and $V_y = 0$ and when both input is 1, no current flows and $V_y = 1$.

For More Material Join: @JEEAdvanced_2025

6) $Q = (B.E)_P - (B.E)_R$
$=(104 + 116)(6.4) - (220)(5.6) = 176$ MeV

7) $\mu_r = 1.6$, $\varepsilon_r = 6.4$
$B = 4.5 \times 10^{-2}$ T
E = ?
$C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$, $V = \frac{1}{\sqrt{\mu \varepsilon}}$
$\frac{C}{V} = \sqrt{\mu_r \varepsilon_r} = \sqrt{1.6 \times 6.4}$
$\frac{E}{B} = V = \frac{3 \times 10^8}{\sqrt{1.6 \times 6.4}} = 9.3 \times 10^7$ m/s.
$E = 4.5 \times 10^{-2} \times 9.3 \times 10^7 = 4.2 \times 10^6$

8) Potential decreases in the direction of E.
 From origin potential decrease from V_0 to certain value
$E = -\frac{dV}{dx}$
 Slope of graph V-x is equal to E
E is high in between the plate
 Slope between plate is maximum.



10) $x_m \propto \frac{1}{T}$
$\frac{(\chi m)_{300K}}{(\chi m)_{350K}} = \frac{350}{300}$
or $(\chi_m)_{300K} = \frac{350}{300} \times 2.8 \times 10^{-4} = 3.267 \times 10^{-4}$

11) $PV^{4/3}$ = constant
$\Rightarrow P^3V^4$ = constant
$\Rightarrow 3nP + 4nV$ = constant
$\Rightarrow 4\frac{dV}{V} = -\frac{3dP}{P}$
$\Rightarrow 4PdV + 3VdP = 0$
$\Rightarrow 3(PdV + VdP) + PdV = 0$
For adiabatic process,
$dQ = dU + dW = 0$
$\Rightarrow dU = -PdV$ ...(i)
$\Rightarrow -PdV = 3(PdV + VdP)$ ...(ii)

For More Material Join: @JEEAdvanced_2025

From equation (i) and (ii)

$\Rightarrow dU = 3(PdV + VdP)$

$\Rightarrow \int dU = 3\int (Pdv + VdP)$

$\Rightarrow U = 3\int d(PV)$

$\Rightarrow U = k + 3PV$

$\Rightarrow n = 3$

12) $mg = 2TL \Rightarrow \pi r^2 Ldg = 2TL \Rightarrow \Rightarrow \pi r^2 dg = 2T.$

This relation is independent of L.

13) Since current is leading the source voltage, this shows that circuit is of capacitive nature.

However, current leads in phase by $\frac{\pi}{4}$, therefore, box also consists of resistance in series with the capacitance.

$\Rightarrow \tan\frac{\pi}{4} = \frac{X_C}{R}$

$\Rightarrow X_C = \frac{1}{\omega C} = R$ .....

$I_{max} = \frac{V_{max}}{\sqrt{R^2 + X_C^2}} = \sqrt{2}A$ .....

16) formula, $I = I_{max}\cos^2 \delta/2$

$\frac{3}{4}I_{max} = I_{max}\cos^2\delta/2 \Rightarrow \delta = \frac{\pi}{3}$

If P is at a distance x from the central maximum.

$\frac{2\pi d}{\lambda D}x = \frac{\pi}{3}$

For the second wavelength,

$\delta' = \frac{2\pi d}{\lambda' D}x = \frac{2\pi d}{\lambda' D} \times \frac{\pi}{3}\frac{D\lambda}{2\pi d} = \frac{\pi}{2}$

$I = I_{max}\cos^2\delta'/2 = \frac{I_{max}}{2}$

17) Use basic concept.

18) From constraint $4a_2 = 3a_1$ .....(i)

Where a_1 and a_2 are acceleration of 6 kg and 4 kg blocks respectively

$4T = 4a_2$ ..... (ii)

$66 - 3T = 6a_1$ .....(iii)

Solving (i), (ii) and (iii) we get

$a_1 = 8\ m/s^2$ and $a_2 = 6\ m/s^2$

19) $\vec{P}_f = \frac{mv_0}{2}\hat{j}$

For More Material Join: @JEEAdvanced_2025

$\vec{P}_i = \frac{4}{5}mv_0\hat{i} + \frac{3}{5}mv_0\hat{j}$

Impulse acting on ball

$\vec{J} = \vec{P}_f - \vec{P}_i$

$\therefore \vec{J} = mv_0\left(-\frac{4}{5}\hat{i} - \frac{1}{10}\hat{j}\right)$

20) with respect to cart $\vec{u} = 20\sqrt{3}\hat{j}$

w.r.t. cart $\vec{a} = -10\sqrt{3}\hat{i} - 10\hat{j}$

minimum radius of curvature

$R_{min} = \frac{(30)^2}{20} = 45\,m$

21) Let ground state energy (in eV) be E,

Then from given condition,

$E_{2n} - E_1 = 204$ eV

$\frac{E_1}{4n^2} - E_1 = 204eV$

$E_1\left(\frac{1}{4n^2} - 1\right) = 204eV$

and $E_{2n} - E_n = 40.8$ eV

$\frac{E_1}{4n^2} - \frac{E_1}{n^2} = 40.8$ eV $\Rightarrow E_1 \times -\frac{3}{4n^2} = 40.8$ eV

From (i) and (ii)

$\Rightarrow n = 2$

Now from equation (ii)

$E_1 = -\frac{4}{3}n^2(40.8)$

Or $E_1 = -217$ eV

and $E_1 = -13.6\ Z^2$

Therefore

$z^2 = \frac{E_1}{-13.6} = \frac{-217.6}{-13.6} = 16$

$\Rightarrow z = 4$

22) $\tau = MB\sin 90^\circ = (\pi r^2 I_1)\ (\mu_0 n I_2)$

$= (\pi r^2 I_1)\ (4\pi \times 10^{-7}\frac{N}{\ell}I_2)$

Here $I_1 = 4A$, $I_2 = 3A$ and $r = 0.01m$

$\tau = 60 \times 10^{-6}$Nm

New Sol.

$\tau = M \times B$

$= 10 \times 4 \times \pi \times 10^{-4}$

$x \times 10^{-6}$

$x\frac{500}{0.4} \times 3$

For More Material Join: @JEEAdvanced_2025

23) $t = nT = 3\left(\frac{2\pi m}{qB}\right) = \frac{6\pi m}{qB}$

$y = \frac{1}{2}\left(\frac{qE}{m}\right)\left(\frac{36\pi^2 m^2}{q^2B^2}\right) = \frac{18\pi^2 E}{\alpha B^2}$

24) $P \propto T^4$, $T \propto \frac{1}{\lambda}$

$\therefore P \propto \frac{1}{\lambda_0^4} \Rightarrow P' = P\left(\frac{64 \times 4}{81}\right)$

25) Field inside the shell is zero, so potential is uniform.

$V = \frac{kq}{2R}$ = 9 kilo volts

26) Formula, $\sum W = \Delta k$

$\because \frac{mg}{2}L - f_k L = 5J$ .....(1)

$-\frac{mg}{2}L - f_k L + FL = 5J$ .....(2)

mgL – FL = 0

$\Rightarrow$ F = mg = 10 N

27) formula, $f = \frac{n}{2L}V$

$\because \frac{f_0 + 5}{f_0 - 5} = \frac{1}{0.95}$

$\Rightarrow f_0 = 195$ Hz

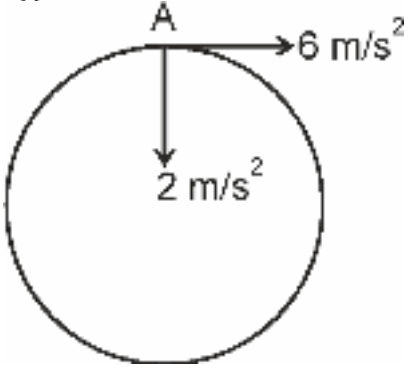
$\because f_0 = \frac{1}{2L}V$

$\Rightarrow L = \frac{V}{2f_0} = \frac{382.2}{2 \times 195} = 0.98m$

28) $\omega = 2$ rad/s

$a_A = \sqrt{6^2 + 2^2}$

$a_A = \sqrt{40}$ m/s^2

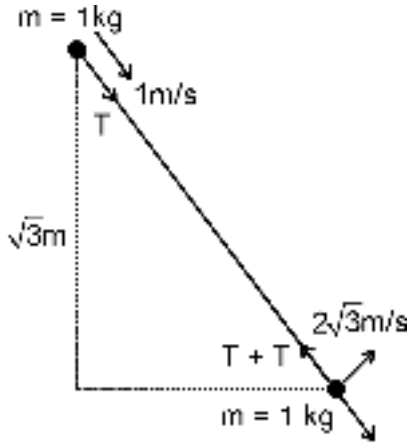


For More Material Join: @JEEAdvanced_2025

29) $p = \frac{(2)^2(4)}{(4-2)^3} = 2$

$\frac{dp}{p} = 2\frac{da}{a} + \frac{db}{b} + 3\frac{dc}{(4-c)}$

$\Rightarrow \frac{dp}{p} \times 100 = 3\%$



30)

w.r.t mass which was at rest initially

$T + \left(\frac{T}{m}\right)m = \frac{mv^2}{\ell}$

$2T = \frac{(1)\left(2\sqrt{3}\right)^2}{2}$

$2T = 6$

$T = 3N$

PART-2 : CHEMISTRY

34) Eq. of NaH_2PO_3 + Eq. of $NaHCO_3$ = Eq. of NaOH

$\frac{20 \times 0.1}{1000} \times 1 + \frac{40 \times 0.1}{1000} \times 1 = x$

$x = 6 \times 10^{-3}$

37) $pH = pK_{a_2} + \log \frac{[CO_3^{2-}]}{[HCO_3^-]}$

$\Rightarrow (11 - \log 4) + \log\left(\frac{1}{2}\right) \Rightarrow 10.1$

39) (A) and (B) statements are correct. These statements are factual.
(C) Along the period, the electronegativity of metals decreases and therefore the difference of electronegativities between metal and oxygen decreases. Hence, the acidic character increases, so the correct order is TiO < VO < CrO < MnO.
(D) V_2O_5 reacts with alkalies as well as acids to give VO_4^{3-} and VO_2^+ respectively.

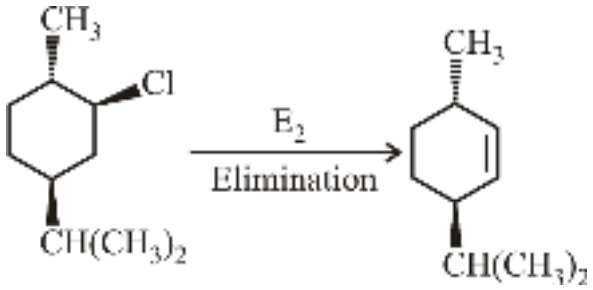
For More Material Join: @JEEAdvanced_2025

41) $2NaNO_3 + (Conc.)H_2SO_4 \rightarrow 2HNO_3 + Na_2SO_4$
$HNO_3 + C \rightarrow H_2O\ (l) + CO_2\ (g) + NO_2(g)$

42) As the electronegativity of central atom decreases bond angle decreases

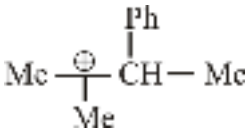
43) This compound has only one ionizable Cl^- out of given three
$[Co(H_2O)_4Cl_2]Cl.2H_2O \rightleftharpoons [Co(H_2O)_4Cl_2]^+ + Cl^-$

44)



45) The pentacetate of glucose does not react with hydroxyl amine indicating the absence of free –CHO group.

46) During reaction

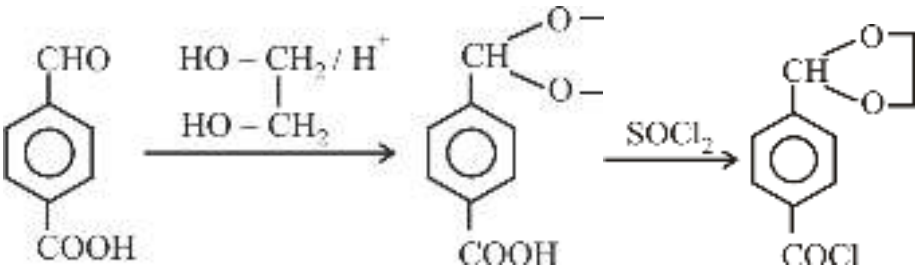


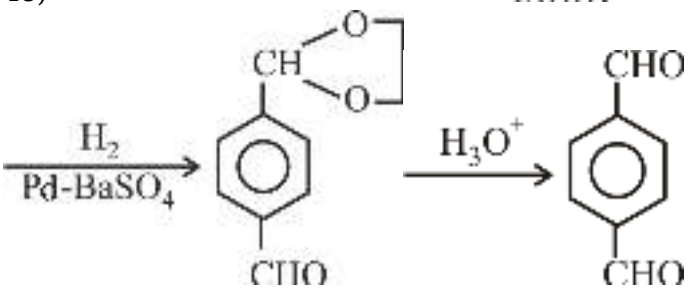
does not form due to unstable carbocation.

47)



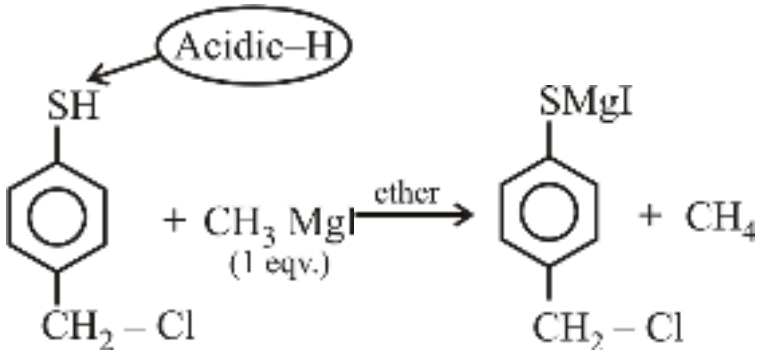
48)





49) Strong E.W.G. not give frictel craft reaction.

For More Material Join: @JEEAdvanced_2025

50) 

[Numerical-value-type question]

51) $\Delta_rH° = -635 - 394 - (-1207) = 178$ kJ/mole

$\Delta_rS° = S°(CaO) + S°(CO_2) - S°(CaCO_3)$

$= 40 + 214 - 94 = 160$ J/K

$\Delta_rG° = \Delta_rH° - T \cdot \Delta_rS° = 178 - \frac{300 \times 160}{1000}$

$= 178 - 48 = 139$ kJ/mole

52) $$E = E^\circ - \frac{0.06}{n} \log \frac{[C_2H_5OH]}{[CH_3CHO]\,[H^+]^2}$$

$$E = 1.76 - \frac{0.06}{n} \log \frac{[5 \times 10^{-10}]}{[5]\,[10^{-1}]^2}$$

$= 1.76 - 0.03 \log 10^{-8}$

$= 1.76 + 0.03 \times 8$

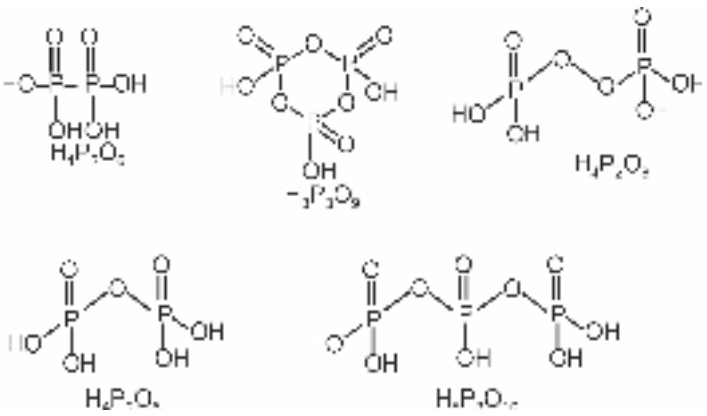
$= 1.76 + 0.24$

E = 2 Volt.

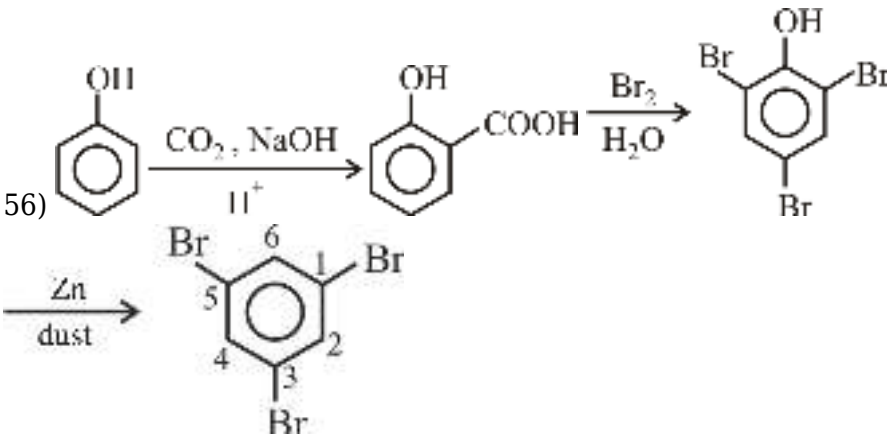
53) $$M = \frac{\text{wt. (NaOH)}}{40} \times \frac{1}{60}$$

Weight (NaOH) $= 40 \times 60 \times 0.5$

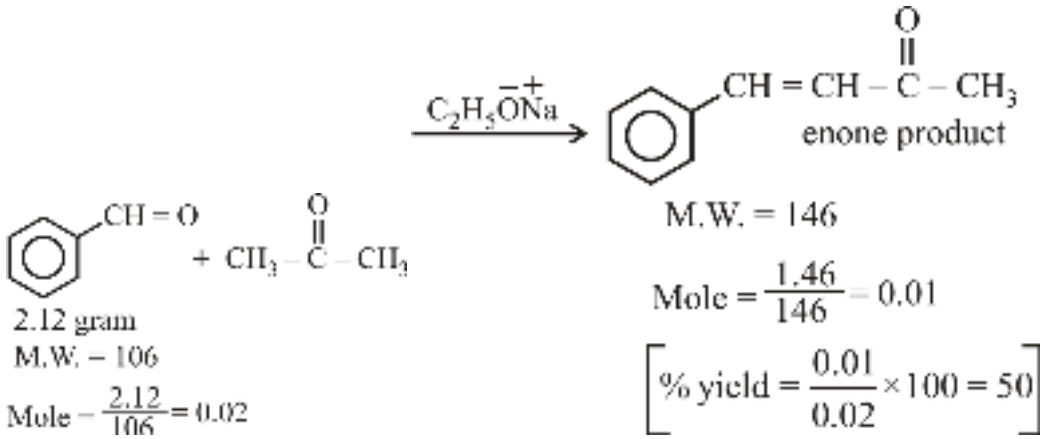
For 80% pure $= \frac{40 \times 60 \times 0.5}{80} \times 100 = 1500$ gm.

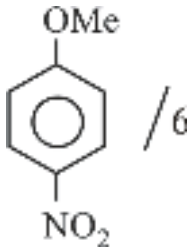
55) 

For More Material Join: @JEEAdvanced_2025

56) 

$\Rightarrow 1 + 3 + 5 = 9$

57) 

58) 

PART-3 : MATHEMATICS

61) 'n' should be multiple of 9 for favourable event. So, $x_1 + x_2 + x_3 = 9$ or 18
If $x_1 + x_2 + x_3 = 9$, $1 \le x_i \le 6$ ...(1)
No. of solution of (1) = coefficient of x^9 in
$(x + x^2 + ... + x^6)^3 = [x(1 - x^6)]^3(1 - x)^{-3}$
= coefficient of x^6 in $(1 - 3x^6)(1 - x)^{-3}$
$= {}^8C_6 + (-3) \times 1 = 25$
And, there is only one solution of
$x_1 + x_2 + x_3 = 18$, $1 \le x_i \le 6$
Hence, n(E) = 26
Also $n(S) = 6^3$
$\Rightarrow P(E) = \frac{13}{108}$

For More Material Join: @JEEAdvanced_2025

62) Let $z = x + iy$
$-8((x + iy)^2 + (x - iy)^2) + 20(x^2 + y^2) = 144$
$-8(2(x^2 - y^2)) + 20(x^2 + y^2) = 144$
$4x^2 + 36y^2 = 144$
$\frac{x^2}{36} + \frac{y^2}{4} = 1$
$a = 6, b = 2$
Area $= \pi ab = 12\pi$

63) Total ways $= 10! - 3.\ 9!\ .\ 2! + 3\ .\ 8! \times 2!\ .\ 2! - 7!\ .\ 2!\ .\ 2!\ .\ 2!$

64) $V = \left[\vec{a} + \left(\vec{a} \times \vec{b}\right) \times \vec{c}\ \ \left(\vec{b} \times \vec{c}\right) \times \vec{a}\ \ \left(\vec{c} \times \vec{a}\right) \times \vec{b}\right]$
$= \left(9\vec{b} - 20\vec{a}\right) . \left[\left(7\vec{c} - 9\vec{b}\right) \times \left(21\vec{a} - 7\vec{c}\right)\right]$
$= 9 \times 7 \times 21 \left[\vec{b}\ \vec{c}\ \vec{a}\right] - 20 \times 9 \times 7 \left[\vec{a}\ \vec{b}\ \vec{c}\right]$
$= 63 \left[\vec{a}\ \vec{b}\ \vec{c}\right] = 63 \sqrt{\begin{vmatrix} 9 & 7 & 9 \\ 7 & 30 & 21 \\ 9 & 21 & 18 \end{vmatrix}} = 63\sqrt{225}$

65) Let normal $y = mx - 2m - m^3$
passing through (3,0)
$\Rightarrow$ $m^3 = m$
$\Rightarrow$ $m = 0, 1, -1$
$\Rightarrow$ P(0,0), Q(1,–2), R(1,2)
$PQ = \sqrt{5}, QR = 4, PR = \sqrt{5}$
$(QR)^2 > (PR)^2 + (PQ)^2 \Rightarrow$ Obtuse angle triangle.

66) $f(x) = (x + 4)(x - 1)(x - 4)|(x - 1)(x + 3)\pi nx|$
Domain of $f(x)$ is $(0, \infty)$

67) Either $x^2 - 2x + 2m$ is perfect square
$m = \frac{1}{2}$ or $x^2 - 3x + m = 0$ and $x^2 - 2x + 2m = 0$ have a common root $\Rightarrow m = -4, 0$

68) (1, 2) & (5, 5) are two focus of hyperbola so
$2ae = 5 \qquad 2a = 3$
$ae = 5/2 \qquad a = 3/2$
$\frac{3}{2}e = \frac{5}{2},\ e = \frac{5}{3}$
and we know that $\frac{1}{e^2} + \frac{1}{e'^2} = 1$
$\frac{1}{e'^2} = 1 - \frac{9}{25} = \frac{16}{25}$
$\frac{1}{e'} = \frac{4}{5} \Rightarrow e' = \frac{5}{4}$

For More Material Join: @JEEAdvanced_2025

$8e' = 8 \times \frac{5}{4} = 10$

69) (A) $(\vec{a}.\vec{c})\,\vec{b} - (\vec{b}.\vec{c})\,\vec{a} = -5\vec{a} + 4\vec{b}$

$\vec{b}.\vec{c} = 5,\ \vec{a}.\vec{c} = 4\ \&\ \vec{a}.\vec{b} = 3$

$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c} = 4\vec{b} - 3\vec{c}$

(B) Parametric equation of line is

$\frac{x-\sqrt{3}}{\cos\theta} = \frac{y}{\sin\theta} = r \Rightarrow \frac{x-\sqrt{3}}{\cos 60^\circ} = \frac{y}{\sin 60^\circ} = r$

$x = \frac{r}{2} + \sqrt{3}\ \&\ y = \frac{\sqrt{3}r}{2}$

$\frac{3r^2}{4} = \frac{r}{2} + 2 + \sqrt{3} \Rightarrow \frac{3r^2}{4} - \frac{r}{2} - \left(2+\sqrt{3}\right) = 0$

$r_1 r_2 = \frac{\left(2+\sqrt{3}\right)}{3/4} = \frac{4\left(2+\sqrt{3}\right)}{3}$

(C) $\sqrt{2}(\cos 2x - \sin x - 1) = 1 + 2\sin x$

$\sqrt{2}(1 - 2\sin^2 x - \sin x - 1) = 1 + 2\sin x$

$-2\sqrt{2}\sin^2 x - \sqrt{2}\sin x = 1 + 2\sin x$

$2\sqrt{2}\sin^2 x + (2+\sqrt{2})\sin x + 1 = 0$

$\sqrt{2}\sin x\,[2\sin x + 1] + 2\sin x + 1 = 0$

$\Rightarrow (\sqrt{2}\sin x + 1)(1 + 2\sin x) = 0$

$\sin x = -1/2$ or $\sin x = -1\sqrt{2}$

$\frac{7\pi}{6}, \frac{11\pi}{6}, \frac{5\pi}{4}, \frac{7\pi}{4}$; sum $= 6\pi$

(D) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{\cos^2 x}dx = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x dx = 2\,[\tan x]_0^{\pi/4} = 2$

70) Let a point on $y^3 = x^4$ be (t^3, t^4)

$3y^2y' = 4x^3$

$\Rightarrow y' = \frac{4x^3}{3y^2} \Rightarrow y' = \frac{4}{3}t$

Equation of tangent is

$y - t^4 = \frac{4t}{3}(x - t^3)$

$\because$ it is a normal to $x^2 + y^2 - 2x = 0$

$\therefore$ it must pass through (1, 0)

$\Rightarrow -\frac{3}{4}t^3 = 1 - t^3 \Rightarrow \frac{t^3}{4} = 1$

$\Rightarrow\ t^3 = 4$

Now $m = \frac{4t}{3}$

$\Rightarrow \left(\frac{3m}{4}\right)^3 = t^3 = 4$

For More Material Join: @JEEAdvanced_2025

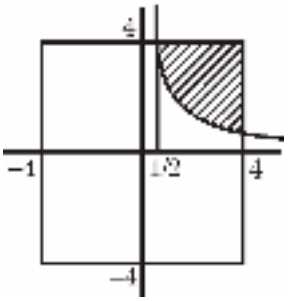
71) $I = \int \frac{dx}{(\sqrt{x})^2 + (\sqrt{x})^7} = \int \frac{dx}{x^{7/2}(1 + x^{-5/2})}$

Now Put $x^{-5/2} = t \Rightarrow \frac{-5}{2x^{7/2}}dx = dt$

$\Rightarrow I = \frac{-2}{5}\int \frac{dt}{1+t} = -\frac{2}{5}\ell n|1+t| + c$

$\Rightarrow \frac{2}{5}\ell n\left|\frac{1}{1+t}\right| + c \Rightarrow \frac{2}{5}\ell n\left|\frac{x^{5/2}}{1+x^{5/2}}\right| + c$

72) $|x| \leq 4$ and $|y| \leq 4$



Area of shaded portion

$\Delta = \int_{1/2}^{4} \frac{2}{x}dx \Rightarrow [2\ell nx]_{1/2}^{4} \Rightarrow 2[\ell n4 + \ell n2] = \ell n64$

Shaded region = 14 – ℓn64

73) $A.M. = \frac{({}^nC_0)^2 + ({}^nC_1)^2 + ... + ({}^nC_n)^2}{(n+1)}$

$= \frac{{}^{2n}C_n}{n+1} = \frac{|2n}{|n\ |n\ (n+1)} = \frac{|2n}{|n\ |n+1}$

74) Let a – 3 = b so that a – 2 = 1 + b.

The given equation is

$\sum_{i=0}^{2n} \lambda_i(1+b)^i = \sum_{i=0}^{2n} \mu_i b^i$

$\Rightarrow \lambda_0 + \lambda_1(1+b) + ... + (1+b)^n + ... + (1+b)^{2n}$

$= \mu_0 + \mu_1 b + ... + \mu_n b^n + ... + \mu_{2n} b^{2n}$,

($\because \lambda_1 = 1\ \forall\ i \geq n$)

Comparing coefficient of b^n we get

${}^nC_n + {}^{n+1}C_n + ...+ {}^{2n}C_n = \mu_n = {}^{2n+1}C_n$.

75) Statement-2 is obviously true

From statement 2, $\left(\frac{a+b+c}{3}\right)^2 \leqslant \frac{a^2+b^2+c^2}{3}$

or $\frac{4d^2}{9} - 2d \leqslant 0$ or $d(2d - 9) \leq 0$

For More Material Join: @JEEAdvanced_2025

$\Rightarrow \quad 0 \leq d \leq \frac{9}{2} \quad \Rightarrow$ Statement 1 is also true.

76) put $x = 0$ and $y = 0 \quad \Rightarrow \quad f^2(0) = 2$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{f^2(x+h)-f^2(x)}{[f(x+h)+f(x)]\,.\,h}$$

$$= \frac{1}{2f(x)} \lim_{h\to 0} \frac{f^2(h)+2(xh-1)}{h}$$

$$= \frac{1}{2f(x)} \lim_{h\to 0} \left[\frac{2xh}{h} + \frac{f^2(h)-2}{h}\right]$$

$$= \frac{1}{2f(x)} \left[2x + \lim_{h\to 0} \frac{f^2(h)-f^2(0)}{h}\right]$$

$$= \frac{1}{2f(x)} \left[2x + \lim_{h\to 0} \frac{f(h)-f(0)}{h} .\ (f(h)+f(0))\right]$$

$$f'(x) = \frac{1}{2f(x)} [2x + 2f(0)\,.\,f'(0)]$$

$\therefore$ $f(x)\,.\,f'(x) = x + f(0).\,f'(0) \Rightarrow f(x)\,.\,f'(x) = x + \lambda$, where $\lambda = f(0) \cdot f'(0)$

integrating both sides

$$\therefore \frac{f^2(x)}{2} = \frac{x^2}{2} + \lambda x + C$$

Integrating both sides

$f^2(x) = x^2 + 2\lambda x + C$

at $x = 0$, $f^2(0) = 2 \quad \Rightarrow \quad C = 2$

at $x = \sqrt{2}$, $f^2(\sqrt{2}) = 4 \quad \Rightarrow \quad \lambda = 0$

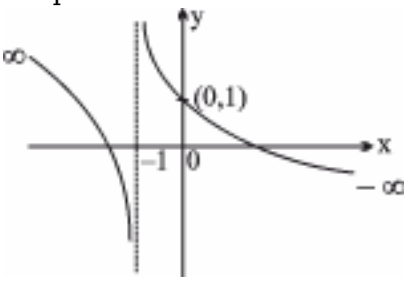
$\therefore \quad f^2(x) = x^2 + 2 \quad \Rightarrow \quad f(x) = \sqrt{x^2+2} \quad (f(x) > 0)$

77) $f'(x) = -\frac{3}{(x+1)^4} - 3 + \cos x < 0$

hence $f(x)$ is always decreasing, Also as $x \to \infty$, $f(x) \to -\infty$ and as $x \to -\infty$, $f(x) \to +\infty$

hence one positive and one negative root

Graph is as shown



78) $\frac{dy}{dx} = \frac{1}{\sqrt{x^4+3x^2+13}}$ when $y = f(x)$

$$\frac{1}{\frac{dy}{dx}} = \sqrt{x^4+3x^2+13}$$

$\therefore \quad g'(y) = \frac{1}{\frac{dy}{dx}}$

when $y = 0$ then $x = 3$

hence $g'(0) = \sqrt{3^4+27+13} = \sqrt{121} = 11$

For More Material Join: @JEEAdvanced_2025

79) n^{th} term of 3, 7, 13, 21, 31 is $n^2 + n + 1$ $T_n = \tan^{-1}\frac{1}{1+n(n+1)} = \tan^{-1}\frac{(n+1)-n}{1+n(n+1)} =$ $\tan^{-1}(n+1) - \tan^{-1} n$

Hence

$S = T_1 + T_2 + T_3 + T_4 + T_5$

$= (\tan^{-1}2 - \tan^{-1}1)$
$(\tan^{-1}3 - \tan^{-1}2)$
$(\tan^{-1}4 - \tan^{-1}3)$
$(\tan^{-1}5 - \tan^{-1}4)$
$(\tan^{-1}6 - \tan^{-1}5)$

$S = \tan^{-1}6 - \tan^{-1}1 = \tan^{-1}\left(\frac{6-1}{1+6}\right) = \tan^{-1}\left(\frac{5}{7}\right)$

$\therefore \tan\left(\tan^{-1}\frac{5}{7}\right) = \frac{5}{7}$

$\square$ integer $= 14 \cdot \frac{5}{7} = 10$

80) $L = \lim_{x\to 0} -\left(\frac{1-\cos x}{x^2}\right)\left(\frac{(e^{2x^2}-1-2x^2)x^2}{x^2 \cdot x^2(\sin 3x - \ln(1+3x))}\right)$

$= -\frac{1}{2}\lim_{t\to 0}\frac{e^{2t}-1-2t}{t^2} \cdot \lim_{x\to 0}\frac{x^2}{\sin 3x - \ln(1+3x)}$ where $x^2 = t$

$= -\frac{1}{2}\lim_{y\to 0}\frac{4(e^y-1-y)}{y^2} \cdot \lim_{x\to 0}\frac{x^2}{\sin 3x - \ln(1+3x)}$

$= -\frac{1}{2}.2.\lim_{x\to 0}\frac{x^2}{\sin 3x - \ln(1+3x)}$

now, $L = \lim_{x\to 0}\frac{x^2}{\sin 3x - \ln(1+3x)}$

using L'Hospital's rule $L = \frac{2}{9}$;

$\square$ $L = -\frac{2}{9}$

81) Let point on line $\left(h, \frac{1}{t+2} - h\right)$

chord of contact $4xh + 16y\left(\frac{1}{t+2} - h\right) = 1$

it passes through $x = 4y$ & $\frac{16y}{t+2} - 1 = 0$

fixed point $= \left(\frac{t+2}{4}, \frac{t+2}{16}\right)$ lies inside ellipse

So $S_1 < 0$

$-\frac{4}{\sqrt{5}} < (t+2) < \frac{4}{\sqrt{5}}$

$t = -1, -3, \quad (t \neq -2)$

For More Material Join: @JEEAdvanced_2025

82) $|A - xI| = 0$

$$\begin{vmatrix} (2-x) & 0 & 1 \\ 1 & -x & 2 \\ 2 & 1 & (1-x) \end{vmatrix} = 0$$

$(2 - x)(x(x - 1) - 2) + 1 + 2x = 0$

$(2 - x)(x^2 - x - 2) + 1 + 2x = 0$

$2x^2 - 2x - 4 - x^3 + x^2 + 2x + 1 + 2x = 0$

$x^3 - 3x^2 - 2x + 3 = 0$

$(A - I)^3 - 5A = -4I$

$|(A - I)^3 - 5A| = -64$

83) Roots of $f(x) = 0$ are, $-3, -2, -1, 0, 1, 2, 3,$

Using rolle's theorem roots of $f'(x) = 0$ lies in $(-3, -2), (-2, -1) \ldots\ldots (2, 3)$

Hence $[\alpha_1] = -3$, $[\alpha_2] = -2 \ldots\ldots [\alpha_6]$

$$= \sum_{i=1}^{6} (-1)^i [\alpha_i] = -(-3) + (-2) - (-1) + 0 - 1 + 2 = 3$$

84) Equation of straight line in parametric form is

$$\frac{x+1}{\cos\theta} = \frac{y-2}{\sin\theta} = r$$

$r = (r\cos\theta) - 1$ & $y = (r\sin\theta) + 2$

satisfied by the curve

$(r\cos\theta - 1)(r\sin\theta + 2) = c^2$

$r^2\sin\theta\cos\theta + r(2\cos\theta - \sin\theta) - 2 - c^2 = 0$

$$r_1 + r_2 = PA + PB = \frac{\sin\theta - 2\cos\theta}{\sin\theta\cos\theta} = 2PQ = 2r_3$$

$2(r_3\sin\theta)(r_3\cos\theta) = (r_3\sin\theta - 2r_3\cos\theta)$

$r_3\cos\theta = h + 1$ & $r_3\sin\theta = k - 2$

$2(h + 1)(k - 2) = (k - 2) - 2(h + 1)$

By solving $y = \dfrac{2x}{(2x+1)}$

85) Let α, β are roots of $x^2 - px + q = 0$ (where α, β are prime) $\alpha + \beta = p$; $\alpha\beta = q$

given $p + q = 11$

$\therefore \quad \alpha + \beta + \alpha\beta = 11$

$(1 + \alpha)(1 + \beta) = 12$

hence $\alpha = 3$ and $\beta = 2$

$\square$ $p = 5$ and $q = 6$

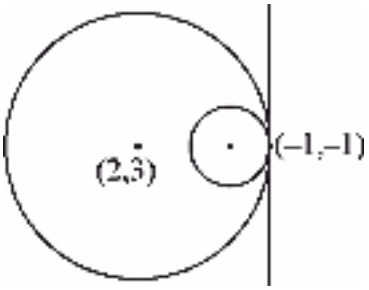
$p^2 + q^2 = 25 + 36 = 61 = a$

$f(a) = f(61) = (61)^2 - 5 \cdot 61 + 6$

$= 61 \cdot 56 + 6 = 3416 + 6 = 3422$

86) $x^2 + y^2 - 4x - 6y - 12 = 0$ ....(1)

For More Material Join: @JEEAdvanced_2025



equation of tangent to (1) at (–1, 1) is

– x – y – 2(x – 1) – 3(y – 1) – 12 = 0

3x + 4y + 7 = 0

required circle can be

$(x + 1)^2 + (y + 1)^2 + \lambda(3x + 4y + 7) = 0$

$x^2 + y^2 + x(2 + 3\lambda) + y(2 + 4\lambda) + 2 + 7\lambda = 0$

$g = \frac{2+3\lambda}{2}$, $f = \frac{2+4\lambda}{2}$ $c = 2 + 7\lambda$

given radius = 3

$\therefore \left(\frac{2+3\lambda}{2}\right)^2 + \left(\frac{2+4\lambda}{2}\right)^2 - 2 - 7\lambda = 9$

$25\lambda^2 - 36 = 0$

$\therefore \quad \lambda^2 = \frac{36}{25}; \quad \therefore \quad \lambda = \pm\frac{6}{5}$

If $\lambda = \frac{6}{5}$ then the circle will have centre on the other side of tangent

$\therefore \quad \lambda = -\frac{6}{5}$;

 circle is $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

a = 5, b = 5, g = –4, f = –7

87) $f(x) = x^3 + ax^2 + bx + c$ $f(-2) = -8 + 4a - 2b + c = -10$

$4a - 2b + c + 2 = 0$ .....(1)

$f'(x) = 3x^2 + 2ax + b$

f '(2/3) = 0

$3 \cdot \frac{4}{9} + 2a\left(\frac{2}{3}\right) + b = 0 \Rightarrow \frac{4}{3} + \frac{4a}{3} + b = 0 \Rightarrow 4 + 4a + 3b = 0$ ....(2)

Also $f(2/3) = \frac{50}{27}$

This gives $\frac{8}{27} + \frac{4a}{9} + \frac{2b}{3} + c = \frac{50}{27}$

$8 + 12a + 18b + 27c = 50$

$12a + 18b + 27c = 42$

$4a + 6b + 9c = 14$ .....(3)

solving (1), (2) and (3) we get

$\boxed{a = -1}$, $\boxed{b = 0}$ and $\boxed{c = 2}$

88) $I = \int_0^{\pi/4} (\pi x - 4x^2) \ln(1 + \tan x) dx$

For More Material Join: @JEEAdvanced_2025

$I = \int_0^{\pi/4} (\pi x - 4x^2)\,[\ln 2 - \ln(1+\tan x)]\,dx$ (using king)

$2I = \ln 2 \int_0^{\pi/4} (\pi x - 4x^2)$

$$I = \frac{\ln 2}{2}\left[\frac{\pi x^2}{2} - \frac{4}{3}x^3\right] = \frac{\ln 2}{2}\left[\frac{\pi^3}{32} - \frac{\pi^3}{48}\right] = \frac{\pi^3}{192}\ln 2$$

89) $\ln\left(x + \sqrt{1+x^2}\right) = g(x)$

$$\Rightarrow g'(x) = \frac{1}{\sqrt{1+x^2}} \quad ....(1)$$

also $\left(f\left(\frac{1}{x}\right)\right)' = f'\left(\frac{1}{x}\right)\cdot\left(-\frac{1}{x^2}\right)$

now $f'(x) = \frac{1}{1+\sqrt{1+x^2}}\cdot\frac{x}{\sqrt{1+x^2}}$

$$f'\left(\frac{1}{x}\right) = \frac{x}{1+\sqrt{1+x^2}}\cdot\frac{1}{x}\cdot\frac{x}{\sqrt{1+x^2}}$$

$$f'\left(\frac{1}{x}\right)\cdot\left(-\frac{1}{x^2}\right) = \frac{1}{x\sqrt{1+x^2}\left(x+\sqrt{1+x^2}\right)} = -\frac{1}{x\sqrt{1+x^2}}\cdot\frac{1}{e^{g(x)}}$$

$$g'(x) + e^{g(x)}\left(f\left(\frac{1}{x}\right)\right)' = 0$$

90) Using $\cos 2\alpha = \frac{1-t_1^2}{1+t_1^2}$ where $t_1 = \tan\alpha$

and $\cos 2\beta = \frac{1-t_2^2}{1+t_2^2}$ where $t_2 = \tan\beta$

we have $\left[1003 - 1002\frac{\left(t-t_1^2\right)}{1+t_1^2}\right]\left[1003 - 1002\frac{\left(t-t_2^2\right)}{1+t_2^2}\right]$

$$= \frac{1003(1+t_1^2) - 1002(1-t_1^2)}{(1+t_1^2)} \times \frac{1003(1+t_2^2) - 1002(1-t_2^2)}{(1+t_2^2)}$$

$$= \frac{(1+2005t_1^2)(1+2005t_2^2)}{1+t_1^2+t_2^2+t_1^2t_2^2}$$ given $t_1t_2 = \frac{1}{\sqrt{2005}} \Rightarrow t_1^2t_2^2 = \frac{1}{2005}$

$$\frac{1+2005(t_1^2+t_2^2)+(2005)^2t_1^2t_2^2}{1+t_1^2+t_2^2+\frac{1}{2005}} = \frac{1+2005(t_1^2+t_2^2)+2005}{1+t_1^2+t_2^2+\frac{1}{2005}}$$

$\Rightarrow$ Hence

$$= \frac{2005\left[\frac{1}{2005}+t_1^2+t_2^2+1\right]}{\left[1+t_1^2+t_2^2+\frac{1}{2005}\right]} = 2005$$

For More Material Join: @JEEAdvanced_2025